

RUPAYAN ROY

Bangalore, KA | +91 9566211693 | rupayan.roy16@gmail.com | linkedin.com/in/rupayan-roy

Professional Summary

Full Stack Software Engineer with 6+ years designing distributed systems and microservices architecture across pharmaceutical and semiconductor industries. Expert in end-to-end development from React frontends to Python/FastAPI backends with proven technical leadership and system design expertise. Strong experience in RESTful API development, real-time systems, event-driven architecture, and cloud infrastructure.

Technical Skills

Languages & Frameworks: Python (FastAPI, Flask), React, TypeScript, JavaScript, C# .NET

Architecture & Design: Microservices, Distributed Systems, RESTful APIs, System Design, Event-Driven Architecture

Databases & Infrastructure: PostgreSQL, Snowflake, Redis, Docker, AWS, Azure

Tools & Practices: Git, CI/CD (GitHub Actions), Agile/Scrum, WebRTC, Code Review, Performance Optimization

Work Experience

The Math Company (MathCo)

Jan 2024 – Present

Senior Product Engineer

Bangalore, India

- Engineered real-time streaming microservices architecture for pharmaceutical competitive intelligence using Server-Sent Events. Implemented EventSource API on React frontend and migrated backend to asynchronous ASGI framework, achieving 80% TTFB reduction (30s to 6s) and 5x concurrency scaling through distributed systems design.
- Led team of 10 engineers defining OpenAPI specs for RESTful API development, conducting code reviews, and enforcing pre-commit hooks. Designed scalable PostgreSQL schemas for complex hierarchical data from PPT, PDF, and Word documents. Architected React frontend with reusable components and feature flags.
- Implemented OIDC authentication with automated role assignment and granular RBAC, ensuring secure access control across distributed services.
- Built real-time voice agent for pharmaceutical sales using GPT-realtime API. Migrated from WebSocket to WebRTC achieving sub-150ms latency per client SLA. Implemented Azure custom NER for PII redaction ensuring HIPAA compliance.
- Refactored FastAPI backend through system design improvements using dependency injection and repository pattern. Implemented event-driven architecture with Redis Streams for async report generation. Improved test coverage from 55% to 85%. Established CI/CD pipeline with GitHub Actions deploying Dockerized microservices to Azure.
- Built mobile-responsive frontend with state management for real-time audio streaming. Led Agile sprint planning, authored feature specs, and approved PRs while enforcing code quality.

Infineon Technologies Ltd.

March 2022 – Dec 2023

Developer - II

Bangalore, India

- Engineered semiconductor configuration web application using MVVM architecture for automotive microchip design. Developed VS Code extension using WebView APIs for custom Meta Description Language, reducing RTL design lead times by %.
- Built Python backend with RESTful APIs using Mako for code generation and parsers, automating hardware configuration workflows.

Infosys Ltd.

Dec 2019 – March 2022

Senior Systems Engineer

Pune, India

- Led MySQL to AWS RDS migration for fintech reconciliation system with zero downtime, reducing failover from 15 minutes to 90 seconds. Migrated CI/CD pipeline from Jenkins to AWS CodePipeline.

Education

SRM Institute of Science and Technology

2015 – 2019

B.Tech in Electronics and Communication Engineering (81%)

Tamil Nadu, India

Awards & Certifications

- 2024 Impact Appreciation Award (MathCo)
- Java Programming: Solving Problems with Software (Coursera)
- Python Data Structures (Coursera)